1. Introduction to PDB Database (2 Marks)

- The **Protein Data Bank (PDB)** is a global repository for the 3D structural data of large biological molecules, such as proteins and nucleic acids.
- **Purpose:** It serves as a central archive for researchers worldwide to deposit, access, and analyze experimental data derived from structural biology techniques. This open access facilitates scientific discovery and understanding of biological processes at a molecular level.
- **Content:** Each entry in the PDB, known as a PDB ID (e.g., 1CRN, 4HHB), represents a unique experimentally determined 3D structure. These structures provide atomic coordinates for every atom in the macromolecule, along with associated metadata.
- **Importance:** Crucial for understanding protein function, enzyme mechanisms, drug design, and disease pathways. It allows for comparative structural analysis and the development of new computational tools.
- **Key Organizations:** The PDB is maintained by the Worldwide Protein Data Bank (wwPDB) consortium, comprising the RCSB PDB (USA), PDBe (Europe), PDBj (Japan), and BMRB (USA, for NMR data).

2. How to Understand and Read 3D Structure Files (2 Marks)

- **PDB File Format:** The primary format for structural data in the PDB is a plain text file with a `.pdb` extension. It's a highly structured text file with specific keywords and column definitions.
- **Key Sections/Records:**
  - **HEADER:** General information like deposition date, classification.
  - **TITLE:** A descriptive title for the entry.
  - **COMPND:** Chemical name of the molecule.
  - **SOURCE:** Biological source of the molecule.
  - **REMARK:** Contains details about the experimental method, resolution (for crystallography), software used, etc. Crucial for assessing data quality.
  - **SEQRES:** The amino acid or nucleotide sequence of the macromolecule.
  - **HETATM:** Coordinates for non-standard atoms, like ligands, cofactors, and water molecules.
  - **ATOM:** The most important section, containing the atomic coordinates (X, Y, Z) for each atom in the protein/nucleic acid, along with atom name, residue name, chain identifier, residue number, occupancy, B-factor (temperature factor), and element symbol.
  - **CONECT:** Information about covalent bonds (though often derived from coordinates for standard residues).
- **Reading/Interpretation:**
  - **Software:** Specialized molecular visualization software (e.g., PyMOL, VMD, Jmol, Chimera) is essential to visualize and interpret the 3D structures from PDB files. These programs convert the coordinate data into a visual representation of atoms and bonds.

- ○ **Understanding Coordinates:** The X, Y, Z values define the position of each atom in 3D space.
- ○ **B-factor (Temperature Factor):** Indicates the displacement of an atom from its average position. Higher B-factors suggest more disorder or flexibility.
- ○ **Occupancy:** Represents the fraction of time an atom occupies a particular position (useful for alternate conformations).

3. Structural Biology Techniques: Basics (3 Marks)

- **a. Macromolecular Crystallography (X-ray Crystallography):**
  - **Principle:** Involves growing highly ordered 3D crystals of a macromolecule. When a beam of X-rays is diffracted by the electrons in the crystal, a diffraction pattern is produced.
  - **Process:**
    1. **Crystallization:** The most challenging step, obtaining high-quality single crystals.
    2. **X-ray Diffraction:** Exposing the crystal to X-rays and collecting diffraction data.
    3. **Phase Problem:** Converting the diffraction pattern (amplitudes) back into an electron density map requires phase information, which is lost during diffraction. Techniques like molecular replacement, MAD/SAD, or MIR are used to solve this.
    4. **Model Building & Refinement:** Building an atomic model into the electron density map and refining it to fit the experimental data.
  - **Output:** High-resolution 3D atomic coordinates.
  - **Limitations:** Requires crystals, can be difficult for flexible molecules.
- **b. Nuclear Magnetic Resonance (NMR) Spectroscopy:**
  - **Principle:** Exploits the magnetic properties of atomic nuclei (e.g., 1H, 13C, 15N) in a strong magnetic field. When exposed to radiofrequency pulses, these nuclei absorb and re-emit energy.
  - **Process:**
    1. **Sample Preparation:** Dissolving the protein in a suitable solvent (usually D_2O or H_2O), often enriched with isotopes.
    2. **Data Acquisition:** Collecting various 1D, 2D, 3D, or even 4D NMR spectra.
    3. **Assignment:** Assigning specific peaks in the spectra to individual atoms in the protein.
    4. **Distance Restraints:** Using NOE (Nuclear Overhauser Effect) to determine inter-atomic distances, which are crucial for defining the 3D structure.

    5. **Structure Calculation:** Using computational methods (e.g., molecular dynamics, simulated annealing) to generate a family of structures that satisfy the experimental distance restraints.
  - **Output:** A ensemble of closely related 3D structures (not a single fixed structure), along with dynamic information.
  - **Limitations:** Best suited for smaller proteins (≤ 50 kDa), requires high protein concentrations.
- **c. Cryo-Electron Microscopy (Cryo-EM):**
  - **Principle:** Involves flash-freezing biological samples in a thin layer of vitreous ice, preserving their native state. High-energy electron beams are then passed through the sample to generate 2D projection images.
  - **Process:**
    1. **Sample Preparation:** Plunging the sample into liquid ethane to vitrify it (rapid freezing to prevent ice crystal formation).
    2. **Imaging:** Collecting thousands of 2D images of individual particles from different orientations using an electron microscope.
    3. **Image Processing:** Selecting good particles, aligning them, and classifying them into groups.
    4. **3D Reconstruction:** Combining the 2D projections into a 3D electron density map using computational algorithms.
    5. **Model Building & Refinement:** Fitting an atomic model into the 3D density map.
  - **Output:** A 3D electron density map, which can then be used to build an atomic model.
  - **Advantages:** Can study large macromolecular complexes, membrane proteins, and heterogeneous samples, does not require crystallization.
  - **Limitations:** Resolution can be lower than crystallography, especially for smaller or more flexible regions.

4. How to Use RCSB PDB (2 Marks)

- **Website:** The primary interface for accessing the PDB is the RCSB PDB website (www.rcsb.org).
- **Searching:**
  - **Simple Search:** Using keywords (e.g., "insulin," "kinase"), PDB ID (e.g., 1CRN), gene name, organism, author.
  - **Advanced Search:** Allows for more complex queries combining multiple criteria (e.g., resolution range, experimental method, ligand presence, binding site features).
- **Navigating a PDB Entry Page:**
  - **Summary Tab:** Provides a quick overview, including resolution, experimental method, deposition date, organism, and key citations.

    - **Macromolecules Tab:** Details about the protein/nucleic acid chains present (sequence, molecular weight, domains).
    - **Ligands Tab:** Information on any bound small molecules (ligands, cofactors, ions).
    - **Structure Tab:** Tools for 3D visualization, downloading files, and viewing molecular dimensions.
    - **Sequence & Annotation Tab:** Sequence alignment, domain information, and links to other databases.
    - **Downloads Tab:** Crucial for downloading the `.pdb` file, `mmCIF` file, electron density maps, and other associated data.
- **Visualization:** Integrated viewers (e.g., Mol* viewer) allow direct 3D visualization within the browser.
- **Tools & Resources:** RCSB PDB also offers various tools for structural analysis, validation, and educational resources.

5. Reading PDB Files and Related Calculations (1 Mark)

- **Reading PDB Files (Programmatically):**
    - While visualization software reads them, for computational analysis, scripts in languages like Python (using libraries like Biopython, MDAnalysis) are often used.
    - The `ATOM` and `HETATM` records are parsed to extract atom names, residue names, chain IDs, residue numbers, and crucially, the X, Y, Z coordinates.
    - Example line from a PDB ATOM record: `ATOM 1 CA ALA A 1 23.456 12.345 34.567 1.00 25.00 C`
        - `ATOM`: Record type
        - `1`: Atom serial number
        - `CA`: Atom name (Alpha Carbon)
        - `ALA`: Residue name
        - `A`: Chain identifier
        - `1`: Residue sequence number
        - `23.456 12.345 34.567`: X, Y, Z coordinates
        - `1.00`: Occupancy
        - `25.00`: B-factor
        - `C`: Element symbol
- **Related Calculations:** Once coordinates are extracted, various calculations can be performed:
    - **Distance Calculation:** Euclidean distance between two atoms using their (X, Y, Z) coordinates: $d=(X_2-X_1)^2+(Y_2-Y_1)^2+(Z_2-Z_1)^2$
    - **Angle Calculation:** Angles between three atoms (e.g., bond angles).
    - **Dihedral/Torsion Angle Calculation:** Angles between four atoms (e.g., phi, psi angles in proteins).

- **Radius of Gyration:** A measure of the compactness of a molecule.
- **Root Mean Square Deviation (RMSD):** Measures the average distance between the atoms of superimposed molecules, used to compare structural similarity.
- **Surface Area/Volume Calculation:** Determining the accessible surface area or volume of a protein.
- **Hydrogen Bond Identification:** Identifying potential hydrogen bonds based on distance and angle criteria.
- **B-factor Analysis:** Analyzing atomic flexibility and disorder.
